DATABASE MANAGEMENT SYSTEMS LAB

Lab Report Submitted to

K. R. Mangalam University

for

Bachelor of Technology

in

Computer Science and Engineering

Submitted by

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LAB ASSIGNMENT- 3

Objectives

- To understand the use of COMMIT and ROLLBACK commands.
- To use ORDER BY and GROUP BY clauses.
- To understand DCL commands (GRANT, REVOKE) and SAVEPOINT.
- To implement JOIN operations in SQL.

CREATE DATABASE

```
--LAB ASSIGNMENT-3
CREATE DATABASE DBMS_LAB3_YUKTI;
USE DBMS_LAB3_YUKTI;
```

CREATE TABLES

DEPARTMENTS (Parent Table)

```
CREATE TABLE DEPARTMENTS (
  dept_no INT PRIMARY KEY,
  dept_name VARCHAR(50),
  dept_location VARCHAR(50)
);
```

TEACHER

```
CREATE TABLE TEACHER (
  Name VARCHAR(50),
  DeptNo INT,
  Date_of_Joining DATE,
  DeptName VARCHAR(50),
  Location VARCHAR(50),
  Salary DECIMAL(10,2)
);
```

SALES

```
CREATE TABLE SALES (
  Sales_No INT PRIMARY KEY,
  Salesname VARCHAR(50),
  Branch VARCHAR(50),
  Salesamount DECIMAL(10,2),
  DOB DATE
);
```

EMPLOYEES (Child Table with Foreign Key)

```
CREATE TABLE EMPLOYEES (
  emp_id INT PRIMARY KEY,
  emp_name VARCHAR(50),
  emp_salary DECIMAL(10,2),
  dept_no INT,
  FOREIGN KEY (dept_no) REFERENCES DEPARTMENTS(dept_no)
);
```

INSERT RECORDS

Insert into DEPARTMENTS

```
INSERT INTO DEPARTMENTS VALUES
(1, 'Mathematics', 'Delhi'),
(2, 'Commerce', 'Mumbai'),
(3, 'Science', 'Chennai'),
(4, 'HR', 'Pune'),
(5, 'IT', 'Bangalore');
```

Insert into TEACHER

```
INSERT INTO TEACHER VALUES
('Amit', 1, '2018-06-12', 'Mathematics', 'Delhi', 50000),
('Neha', 2, '2019-04-10', 'Commerce', 'Mumbai', 55000),
('Ravi', 1, '2020-08-01', 'Mathematics', 'Delhi', 48000),
('Priya', 3, '2017-01-15', 'Science', 'Chennai', 60000),
('Karan', 2, '2021-09-20', 'Commerce', 'Mumbai', 52000);
```

Insert into SALES

```
INSERT INTO SALES VALUES
(101, 'Rahul', 'North', 20000, '1995-12-21'),
(102, 'Simran', 'South', 30000, '1992-11-15'),
(103, 'Ankit', 'North', 25000, '1998-12-05'),
(104, 'Pooja', 'East', 40000, '1994-07-10'),
(105, 'Vikas', 'South', 35000, '1993-12-30');
```

Insert into EMPLOYEES

```
INSERT INTO EMPLOYEES VALUES
(1, 'Arjun', 45000, 1),
(2, 'Meera', 55000, 2),
(3, 'Kabir', 60000, 3),
(4, 'Sneha', 50000, 2),
(5, 'Rohan', 70000, 5);
```

DML OPERATIONS

TEACHER Updates

25% increment for Mathematics

```
UPDATE TEACHER
SET Salary = Salary + (Salary * 0.25)
WHERE DeptName = 'Mathematics';
```

15% increment for Commerce

```
UPDATE TEACHER
SET Salary = Salary + (Salary * 0.15)
WHERE DeptName = 'Commerce';
```

SALES Queries
Total Sales Per Branch

```
SELECT Branch, SUM(Salesamount) AS Total_Sales
FROM SALES
GROUP BY Branch;
```

Branch	Total_Sales
North	45000.00
South	65000.00
East	40000.00

Average Sales Per Branch

```
SELECT Branch, AVG(Salesamount) AS Average_Sales
FROM SALES
GROUP BY Branch;
```

Branch	Average_Sales
North	22500.000000
South	32500.000000
East	40000.000000

Born in December (Formatted)

```
SELECT Salesname,
DATE_FORMAT(DOB, '%d-%b-%y') AS DOB
FROM SALES
WHERE MONTH(DOB) = 12;
```

	Salesname	DOB
▶	Rahul	21-Dec-95
	Ankit	05-Dec-98
	Vikas	30-Dec-93

Alphabetical Order of Month

```
SELECT Salesname, DOB
FROM SALES
ORDER BY MONTHNAME(DOB);
```

	Salesname	DOB
▶	Rahul	1995-12-21
	Ankit	1998-12-05
	Vikas	1993-12-30
	Pooja	1994-07-10
	Simran	1992-11-15

JOIN OPERATIONS

Inner Join

```
SELECT *
FROM EMPLOYEES E
JOIN DEPARTMENTS D
ON E.dept_no = D.dept_no;
```

	Salesname	DOB
▶	Rahul	1995-12-21
	Ankit	1998-12-05
	Vikas	1993-12-30
	Pooja	1994-07-10
	Simran	1992-11-15

LEFT OUTER JOIN

```
SELECT E.emp_name, D.dept_name
FROM EMPLOYEES E
LEFT JOIN DEPARTMENTS D
ON E.dept_no = D.dept_no;
```

emp_id	emp_name	emp_salary	dept_no	dept_no	dept_name	dept_location
1	Arjun	45000.00	1	1	Mathematics	Delhi
2	Meera	55000.00	2	2	Commerce	Mumbai
3	Kabir	60000.00	3	3	Science	Chennai
4	Sneha	50000.00	2	2	Commerce	Mumbai
5	Rohan	70000.00	5	5	IT	Bangalore

RIGHT OUTER JOIN

```
SELECT E.emp_name, D.dept_name
FROM EMPLOYEES E
RIGHT JOIN DEPARTMENTS D
ON E.dept_no = D.dept_no;
```

emp_name	dept_name
Arjun	Mathematics
Meera	Commerce
Kabir	Science
Sneha	Commerce
Rohan	IT

Salary > Average Salary

```
SELECT *
FROM EMPLOYEES
WHERE emp_salary > (SELECT AVG(emp_salary) FROM EMPLOYEES);
```

emp_id	emp_name	emp_salary	dept_no
3	Kabir	60000.00	3
5	Rohan	70000.00	5
NULL	NULL	NULL	NULL

TCL (Transaction Control)

ROLLBACK Example

```
START TRANSACTION;

UPDATE TEACHER
SET Salary = Salary + 5000
WHERE Name = 'Amit';

ROLLBACK;
```

COMMIT Example

```
START TRANSACTION;

UPDATE TEACHER
SET Salary = Salary + 5000
WHERE Name = 'Amit';

COMMIT;
```

SAVEPOINT Example

```
START TRANSACTION;
UPDATE TEACHER SET Salary = Salary + 2000 WHERE Name='Neha';
SAVEPOINT sp1;
UPDATE TEACHER SET Salary = Salary + 3000 WHERE Name='Ravi';
ROLLBACK TO sp1;
COMMIT;
```

DCL COMMANDS

Grant All Privileges

```
GRANT ALL PRIVILEGES ON EMPLOYEES TO DEPARTMENTS;
```

Grant Selected Privileges

```
GRANT SELECT, INSERT ON EMPLOYEES TO DEPARTMENTS;
```

Revoke All Privileges

REVOKE ALL PRIVILEGES ON EMPLOYEES FROM DEPARTMENTS;

Revoke Selected Privileges

REVOKE SELECT, INSERT ON EMPLOYEES FROM DEPARTMENTS;

RESULT

Thus, the required tables were created and records were inserted successfully. DDL, DML, JOIN, TCL (COMMIT, ROLLBACK, SAVEPOINT), and DCL (GRANT, REVOKE) commands were executed and verified successfully.